

Week 4: Lectures Notes

Textbook: Laura Taalman, Peter Kohn: CALCULUS, *Single Variable, 1e*

LIMITS

Limiting behavior occurs in a variety of practical situations:

(a) An area of a disk is the limit of the sum of areas of triangles inscribed in the circle:

(b) In Physics, the temperature 0°K (absolute zero) at which all-molecular activity ceases can be approached but never actually attained in practice. Similarly, economics speaks of profit under ideal conditions, etc.

(c) In mathematics the terms of the sequence $\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots, \frac{1}{2^n}, \dots$ get smaller and smaller with increasing n but never become equal to 0.

Limits of series

Computing the sums from the previous infinite sequence: $\frac{1}{2}, \frac{1}{2} + \frac{1}{4}, \frac{1}{2} + \frac{1}{4} + \frac{1}{8}, \dots$

we obtain the sequence of sums equal to: $\frac{1}{2}, \frac{3}{4}, \frac{7}{8}, \dots, \frac{2^k - 1}{2^k}, \dots, \frac{1023}{1024}, \dots$

The more terms in the series the closer the number to 1.

Increasing the number of terms, we can get closer and closer to 1.

Greek philosopher Zeno of Elea (ca. 490–430 BC) had devised a Dichotomy Paradox: tortoise travelling a certain distance to a pond will never reach the pond if it first covers a half of the distance ($\frac{D}{2}$), then a half of the remaining distance ($\frac{D}{4}$) then half of remaining distance again and again.

The resolution of the paradox is in property of Geometric Series.

$\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2^n} + \dots = \sum_{i=1}^{\infty} \left(\frac{1}{2}\right)^i = 1$ and in the fact that infinite number of “half-steps” needed is balanced by increasingly short amount of time needed to cover the distance.

Limits of average rate of change.

If you drive a car and cover the distance $S=100$ km in 2h, then the average speed is $\bar{v} = 50\text{km} / \text{h}$ and it's mathematically defined as $\bar{v} = \frac{S(2) - S(0)}{2} = \frac{\Delta S(t)}{\Delta t}$.

We can also find an average velocity during smaller time intervals $\bar{v} = \frac{S(1) - S(0)}{1}, \frac{S(0.5) - S(0)}{0.5}, \dots$

It is clear that in order to estimate the velocity of the car in any instant of time (*instantaneous velocity*), we need to approach $\Delta t \rightarrow 0$. In other words,

the *limit of the ratio* $\frac{\Delta S(t)}{\Delta t}$ when $\Delta t \rightarrow 0$ is the instantaneous velocity of your car. In other words,

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Limits that show us an instantaneous rate of change of quantity are called derivatives

Limits of partial sums (area under the curve)

Consider the area between the graph of $f(x)$ and x-axis from $x = 0$ to $x = 1$.

Clearly as we left the number N of rectangles get larger and larger, the sum of the areas of rectangles gets closer and closer to the actual area under the graph of $f(x)$.

$$A_4 = \frac{1}{4}f\left(\frac{1}{4}\right) + \frac{1}{2}f\left(\frac{1}{2}\right) + \frac{1}{4}f\left(\frac{3}{4}\right) + \frac{1}{4}f(1) = 0.4688$$

$$A_8 = \frac{1}{8}f\left(\frac{1}{8}\right) + \frac{1}{8}f\left(\frac{2}{8}\right) + \dots + \frac{1}{8}f(1) = 0.3984$$

Increasing the number of subintervals we can make the area to be close and close to the actual area under the parabola A . We can say that the area under the curve is equal to the limit of the sum

$$\sum_{i=1}^n \frac{1}{n} f\left(\frac{i}{n}\right) \text{ when the "length of the intervals approach to 0" or}$$

"the number of intervals $n \rightarrow \infty$ " (Note: infinity is not a number it's a symbol that we use to show the result of division by 0).

Limits of Functions.

In short, “terms of the sequence approach to 0 when $n \rightarrow \infty$ ”, “limit of the sequence of sums is equal to 1 when $k \rightarrow \infty$ ”, “average velocity of the car is equal to its instantaneous velocity when $\Delta t \rightarrow 0$ ”, can be generalized as:

“the value of a function approaches some number L as its argument gets closer and closer to some value c ”. In calculus this type of behavior is expressed by writing $\lim_{x \rightarrow c} f(x) = L$

Geometrically, the limit statement $\lim_{x \rightarrow c} f(x) = L$ means that the height of the graph $y = f(x)$ approaches L as x approaches c . We can say that $\lim_{x \rightarrow c} f(x) = L$ means that we can make $f(x)$ as close as we want to L by taking x as close as we need to c . (Intuitive definition)

4.1 Informal Definition of Limit

Let's investigate the behavior of function $f(x) = \frac{\sin x}{x}$ for values of x near 0. Note that $f(x)$ is not defined at $x = 0$.

$x > 0$	$f(x)$	$x < 0$	$f(x)$
0.1	0.99833	-0.1	0.99833
0.01	0.9999833	-0.01	0.9999833
0.001	0.99999833	-0.001	0.99999833

Table shows that $f(x)$ gets closer and closer to a number $L = 1$, as x gets closer and closer to $c = 0$ from both sides.

In other words, $f(x)$ can be made **arbitrarily close to 1 whenever x is sufficiently close to 0**. Mathematically this type of behavior can be expressed as $\lim_{x \rightarrow 0} f(x) = 1$

Limit of a function at a point.

“as close as we need” $\rightarrow$ “arbitrarily close”; “as close as we want” $\rightarrow$ “sufficiently close”

DEFINITION 1. Informal Definition of Limit

The limit of $f(x)$ as x approaches c , is equal to L means that $f(x)$ can be made arbitrarily close to L whenever x is sufficiently close to c (but not equal c). We denote this by

$$\lim_{x \rightarrow c} f(x) = L \text{ or } f(x) \rightarrow L \text{ as } x \rightarrow c$$

If limit L is a finite number we say that limit exists and $f(x)$ converges to L

Function is defined at $x=c$.
Its value $f(c) = L$ $f(c) = L$.

Function is not defined at $x=c$

Function is defined at $x=c$,
but its value $f(c) \neq L$

One-Sided Limits

Let's investigate behavior of $f(x) = \frac{\sqrt{9x^2 - 36x + 36}}{x - 2}$ near $x = 2$

$x < 2$	$f(x)$	$x > 2$	$f(x)$
1.9	-3	2.1	3
1.99	-3	2.01	3
1.999	-3	2.001	3
1.9999	-3	2.0001	3

When x approaches 2 from the left side, $f(x)$ approaches -3

When x approaches 2 from the right side $f(x)$ approaches 3

When we say " x approaches c " we mean that x approaches c from the either side.
Note: notation c^- or c^+ *does not mean* that c is negative or positive.

DEFINITION 2

Left - Hand Limit of $f(x)$ is equal to L ($\lim_{x \rightarrow c^-} f(x) = L$) means that we can make the value of $f(x)$ arbitrary close to L by taking x sufficiently close to c , *but less* than c .
(x approaches c from the left side)

DEFINITION 3

Right - Hand Limit of $f(x)$ is equal to L ($\lim_{x \rightarrow c^+} f(x) = L$) means that we can make the value of $f(x)$ arbitrary close to L by taking x sufficiently close to c , *but greater* than c .
(x approaches c from the right side)

Infinite limits

Function $f(x) = \frac{1}{x^2}$ is not defined at 0 and its graph reveals that $f(x)$ increases without bound as $x \rightarrow 0$.

The fact that the function has no numerical limit and becomes arbitrary large is expressed as $\lim_{x \rightarrow 0^-} \frac{1}{x^2} = \infty$ and $\lim_{x \rightarrow 0^+} \frac{1}{x^2} = \infty$. Both one-sided limits are equal and we can say that $\lim_{x \rightarrow 0} \frac{1}{x^2} = \infty$.

DEFINITION 4. Infinite Limit

Let $f(x)$ be a function defined on both sides of c , except possibly at c .

$\lim_{x \rightarrow c} f(x) = \infty$ means that $f(x)$ can take arbitrary large values (as large as we want) when x is sufficiently close to c , but not equal to c .

DEFINITION 5 Vertical asymptote.

Function $f(x)$ has a vertical asymptote at $x = c$ if one or more of the following is true:

$$\lim_{x \rightarrow c^-} f(x) = \infty ; \quad \lim_{x \rightarrow c^+} f(x) = -\infty ; \quad \lim_{x \rightarrow c^-} f(x) = -\infty ; \quad \lim_{x \rightarrow c^+} f(x) = \infty$$

$$\lim_{x \rightarrow 1} f(x) = \infty$$

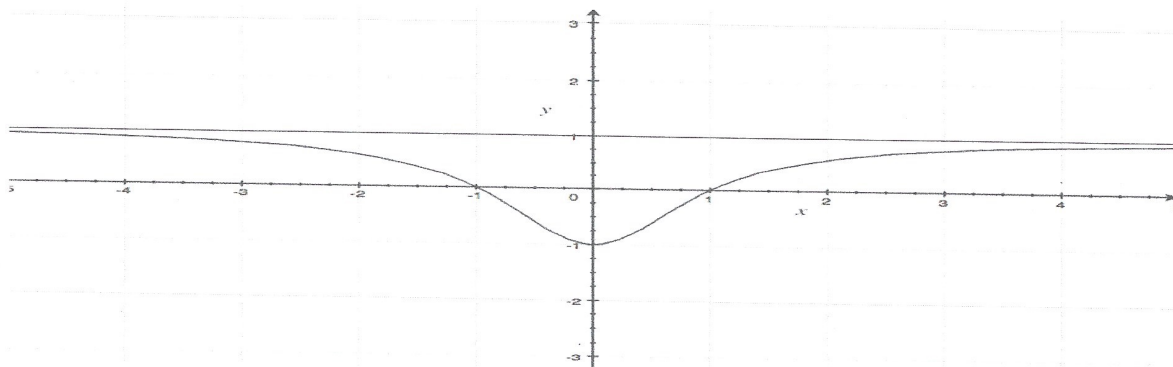
$$\lim_{x \rightarrow 1^-} f(x) = -\infty;$$

$$\lim_{x \rightarrow 1^+} f(x) = \infty$$

Limits at infinity.

Let's investigate the behavior of function $f(x) = \frac{x^2 - 1}{x^2 + 1}$ as x becomes arbitrary large.

x	0	± 1	± 2	± 3	± 4	± 5	± 10	± 50	± 100	± 1000
$f(x)$	-1	0	0.600	0.800	0.8800	0.9230	0.9801	0.9992	0.9998	0.9999



As x grows larger and larger you can see that the values of $f(x)$ get closer and closer to 1. In other words we can make the values of $f(x)$ as close as we like to 1 by taking x sufficiently large. This situation is expressed symbolically by writing $\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = 1$. In general we use notation $\lim_{x \rightarrow \infty} f(x) = L$ to indicate that values of $f(x)$ approach L as x increases without bound.

DEFINITION 6 Limits at infinity.

If values of a function approach L as x growth without bound, then we say that L is the limit of f as x approaches to $\pm\infty$. **Note:** It is also can be said that limit at infinity is equal to L .
 If f is defined on an interval (a, ∞) , then $\lim_{x \rightarrow \infty} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L for all sufficiently large positive x .
 If f is defined on an interval $(-\infty, a)$, then $\lim_{x \rightarrow -\infty} f(x) = L$ means that $f(x)$ can be made arbitrarily close to L for all sufficiently large negative x .

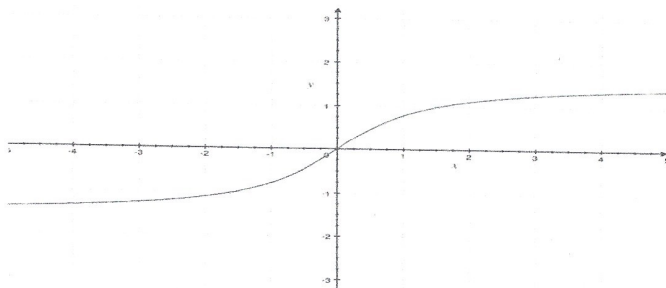
DEFINITION 7 Horizontal asymptote.

A nonconstant function f has a **horizontal asymptote** at $y = L$ if one or both of the following is true: $\lim_{x \rightarrow \infty} f(x) = L$ or $\lim_{x \rightarrow -\infty} f(x) = L$

For example the curve $f(x) = \frac{x^2 - 1}{x^2 + 1}$ has line $y = 1$ as a horizontal asymptote because

$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{x^2 + 1} = 1$. The graph of the function $y = \arctan x$ has two horizontal asymptotes $y = -\frac{\pi}{2}$ and

$y = +\frac{\pi}{2}$ because $\lim_{x \rightarrow \infty} \arctan x = \frac{\pi}{2}$ and $\lim_{x \rightarrow -\infty} \arctan x = -\frac{\pi}{2}$



Limit that does not exist.

1. $\lim_{x \rightarrow 0} \left(\sin \frac{\pi}{x} \right)$

The graph of $y = \sin \frac{\pi}{x}$ shows that the values of the function oscillate infinitely often between -1 and +1 as $x \rightarrow 0$.

We can see that argument of the sine function goes to positive or negative infinity and the values of the function oscillate between -1 and +1. Therefore $\lim_{x \rightarrow 0} \left(\sin \frac{\pi}{x} \right)$ continues to oscillate between +1 and -1 as $x \rightarrow 0$. The behavior exhibited in this example is called **divergence by oscillation** and we can conclude that $\lim_{x \rightarrow 0} \left(\sin \frac{\pi}{x} \right)$ DNE.

Intervals in $(\delta - \varepsilon)$ terms.

EXAMPLES.

Find a value of δ that satisfies the given statements.

a) There exists $\delta > 0$ such that for all x : if x is within distance δ of 2 then $5x+1$ is within distance 1 of 7.

b) There exists $\delta > 0$ such that for all x : if $|x-2| < \delta$ then $|3x-6| < 0.3$.

c) For all $\varepsilon > 0$ there exists $\delta > 0$ such that if x is within distance δ of 3, then $2x$ is within distance ε of 6.

4.2 Formal Definition of Limit

The intuitive definition of a limit, given in previous section is inadequate in mathematical sense because such phrases as “ x is close to 0” or “ $f(x)$ gets closer and closer to L ”.

In order to be able to prove conclusively that $\lim_{x \rightarrow 0} \left(\frac{\sin x}{x} \right) = 1$ or $\lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} = e$

we must make the definition of limit precise.

Assumptions:

1.

$f(x)$ is defined on *punctured* interval $(c - \delta, c) \cup (c, c + \delta)$ where δ represents *infinitesimally* small distance to the left and to the right of $x = c$ on x -axis.

2.

to mathematically precisely express “ $f(x)$ approaches L ” we’ll introduce interval $(L - \varepsilon, L + \varepsilon)$ on the y -axis

3.

in order to guarantee, that $f(x)$ *within distance* $\varepsilon > 0$ *from* L
we must choose values of x *within distance* $\delta > 0$ *from* c , $x \neq c$

DEFINITION 8. **Formal Definition of Limit (Geometric)**

Let $f(x)$ be a function defined on some open interval that contains the number c , except possibly c itself. Then we say that $\lim_{x \rightarrow c} f(x) = L$ if

Definition 8 can be restated as follows:

DEFINITION 8*

$\lim_{x \rightarrow c} f(x) = L$ means that for every $\varepsilon > 0$ we can find $\delta > 0$ such that if x in the open interval $(c - \delta, c + \delta)$ and $x \neq c$, then $f(x)$ lies in the open interval $(L - \varepsilon, L + \varepsilon)$

DEFINITION 8**

If any small interval $(L - \varepsilon, L + \varepsilon)$ is given around L , then we can find an interval $(c - \delta, c + \delta)$ around c such that maps all the points in $(c - \delta, c + \delta)$ except possibly c into the interval $(L - \varepsilon, L + \varepsilon)$.

Example: Write the following limit as a formal statement involving δ and ε : $\lim_{x \rightarrow 3} (x^2 - 4) = 5$

Solution

THEOREM 4.2.1 Uniqueness of Limits.

If $\lim_{x \rightarrow c} f(x) = L$ and $\lim_{x \rightarrow c} f(x) = M$ then $L=M$.

Proof

DEFINITION 9 **One-Sided Limits.**

The **left limit** $\lim_{x \rightarrow c^-} f(x) = L$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in (c - \delta, c)$, then $f(x) \in (L - \varepsilon, L + \varepsilon)$

The **right limit** $\lim_{x \rightarrow c^+} f(x) = L$ means that for all $\varepsilon > 0$, there exists $\delta > 0$ such that if $x \in (c, c + \delta)$, then $f(x) \in (L - \varepsilon, L + \varepsilon)$

THEOREM 4.2.2

$$\lim_{x \rightarrow c} f(x) = L \text{ if and only if } \lim_{x \rightarrow c^-} f(x) = \lim_{x \rightarrow c^+} f(x) = L$$

Proof

QED

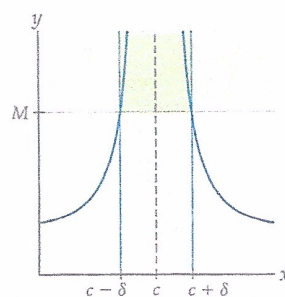
DEFINITION 10. **Infinite Limits and Limits at Infinity.**

The **infinite limit** $\lim_{x \rightarrow c} f(x) = \infty$ means that for all $M > 0$, there exists $\delta > 0$ such that if $x \in (c - \delta, c) \cup (c + \delta, c)$, then $f(x) \in (M, \infty)$

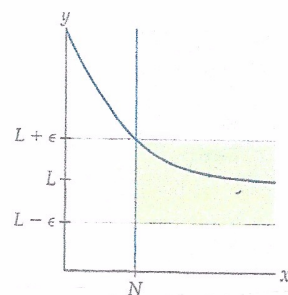
The **limit at infinity** $\lim_{x \rightarrow \infty} f(x) = L$ means that for all $\varepsilon > 0$, there exists $N > 0$ such that if $x \in (N, \infty)$, then $f(x) \in (L - \varepsilon, L + \varepsilon)$

The **infinite limit at infinity** $\lim_{x \rightarrow \infty} f(x) = \infty$ means that for all $M > 0$, there exists $N > 0$ such that if $x \in (N, \infty)$, then $f(x) \in (M, \infty)$

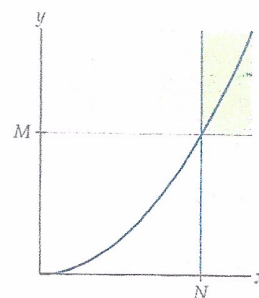
$$\lim_{x \rightarrow c} f(x) = \infty$$



$$\lim_{x \rightarrow \infty} f(x) = L$$



$$\lim_{x \rightarrow \infty} f(x) = \infty$$



Example: Write the following limit as a formal statement involving δ and M : $\lim_{x \rightarrow 2} \frac{4}{(x-2)^2} = \infty$

Solution:

EXAMPLES

1. Given that $\lim_{x \rightarrow 2} (2x + 1) = 5$. Find the **largest** value of δ such that if $\varepsilon = 0.01$ and $x \in (2 - \delta, 2) \cup (2, 2 + \delta)$ then $f(x) \in (5 - \varepsilon, 5 + \varepsilon)$

Solution

2. Show that $\lim_{x \rightarrow 2} (2x + 1) \neq 4$ By finding $\varepsilon > 0$ for which there is no corresponding $\delta > 0$ satisfying the formal definition of limit.

Solution

3. If $x \in (1.5, 2.5)$ what is the **smallest** interval $I = (4 - \varepsilon, 4 + \varepsilon)$ for which we can guarantee that $x^2 \in I$?

4. Write $\lim_{x \rightarrow \infty} f(x) = -\infty$ as a formal statement involving M, N . Sketch the graph that illustrates the role of these constants.

Solution.

5. For $\lim_{x \rightarrow 5} \sqrt{x-1} = 2$, $\varepsilon = 0.2$ approximate the **largest** value of δ such that if $x \in (c - \delta, c) \cup (c + \delta, c)$ then $f(x) \in (L - \varepsilon, L + \varepsilon)$

6. For $\lim_{x \rightarrow 1^+} \frac{1}{x^2 - 1} = \infty$, $M = 1000$ find the **largest** value of $\delta > 0$ that satisfies the given statement.

7. For $\lim_{x \rightarrow 1^+} \frac{3x}{x+1} = 3$, $\varepsilon = 0.1$ find the **smallest** value of $\delta > 0$.

8. Use Geometric definition of Limit to prove that $\lim_{x \rightarrow 2} (7 - x) = 5$

Solution

- a) What is the $\delta - \varepsilon$ statement that must be shown to prove that $\lim_{x \rightarrow 2} (7 - x) = 5$.

Answer:

- b) Argue that $x \in (2 - \delta, 2) \cup (2, 2 + \delta)$ iff $-\delta < x - 2 < \delta$, then use algebra to show that this means that $0 < |x - 2| < \delta$

Answer:

- c) Argue that $7 - x \in (5 - \varepsilon, 5 + \varepsilon)$ iff $-\varepsilon < x - 2 < \varepsilon$, then use algebra to show that this means that $|x - 2| < \varepsilon$

Answer:

- d) Given any particular $\varepsilon > 0$. What value of δ would guarantee that if $0 < |x - 2| < \delta$ then $|x - 2| < \varepsilon$?

Answer:

- e) For given $\varepsilon > 0$ and $\varepsilon = \delta$. If $x \in (2 - \delta, 2) \cup (2, 2 + \delta)$ then $7 - x \in (5 - \varepsilon, 5 + \varepsilon)$
This statement is the formal definition of $\lim_{x \rightarrow 2} (7 - x) = 5$

QED